TLS-PSK Handshakes and the UIM

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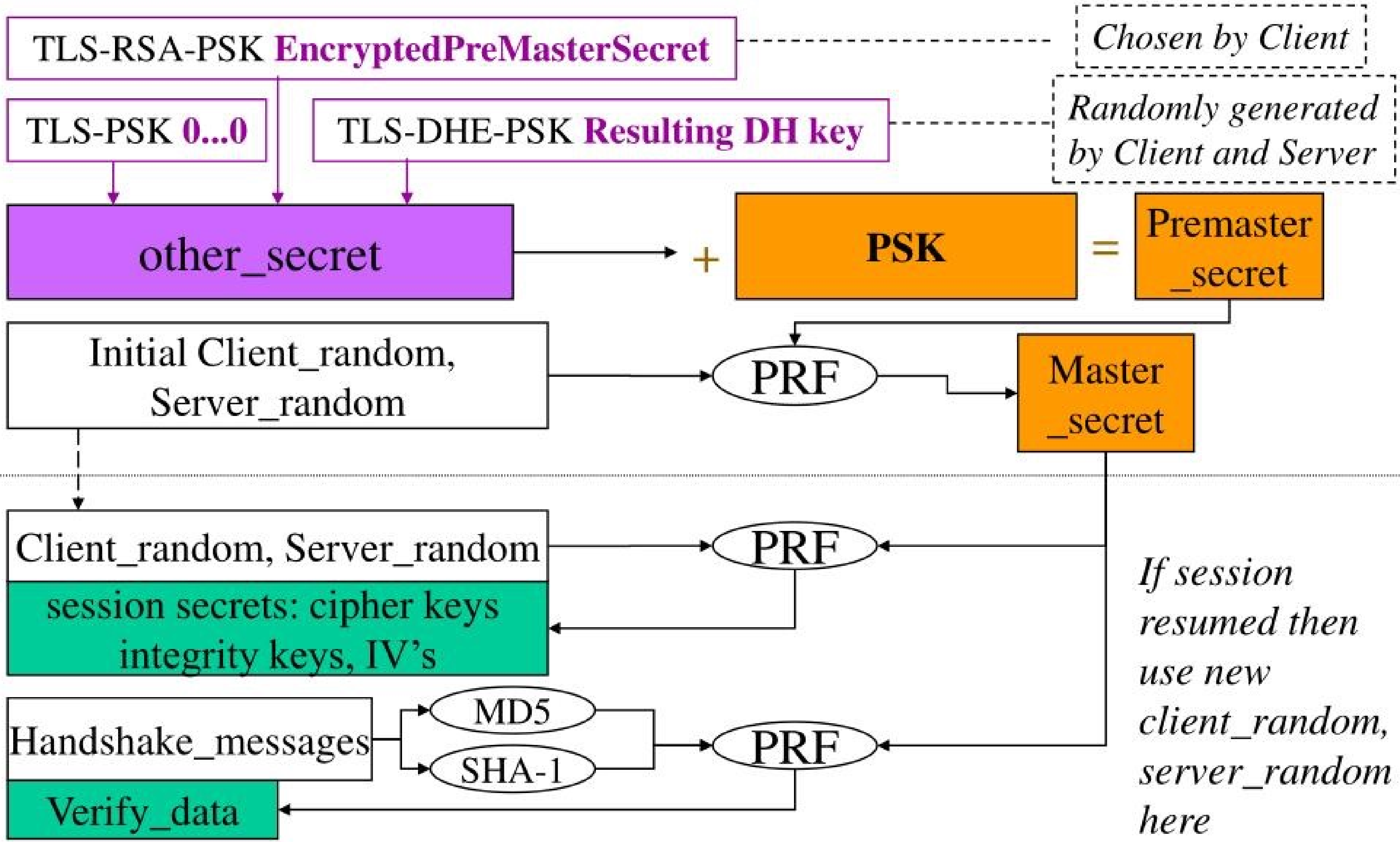
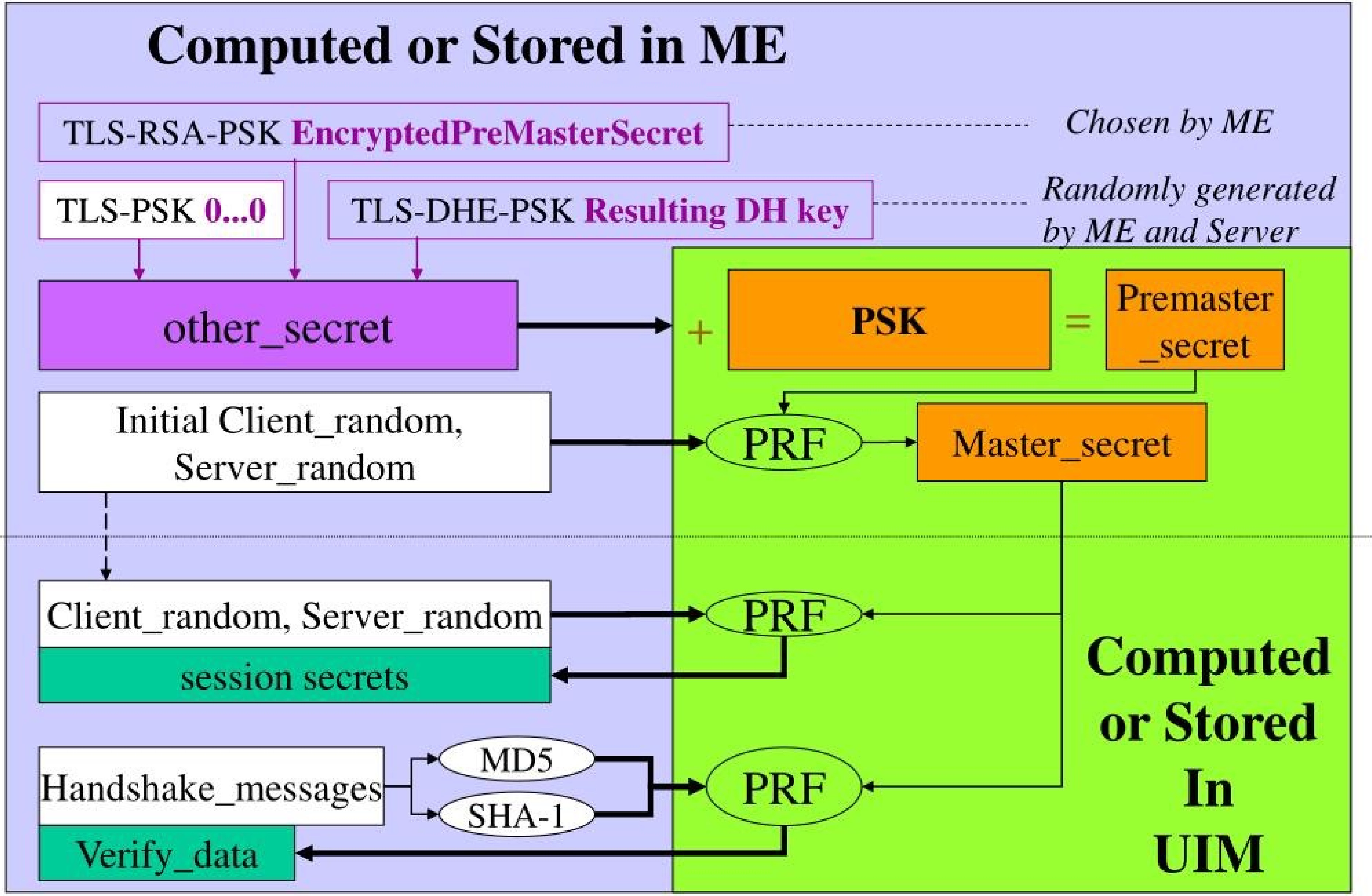
TLS-PSK Key Generation



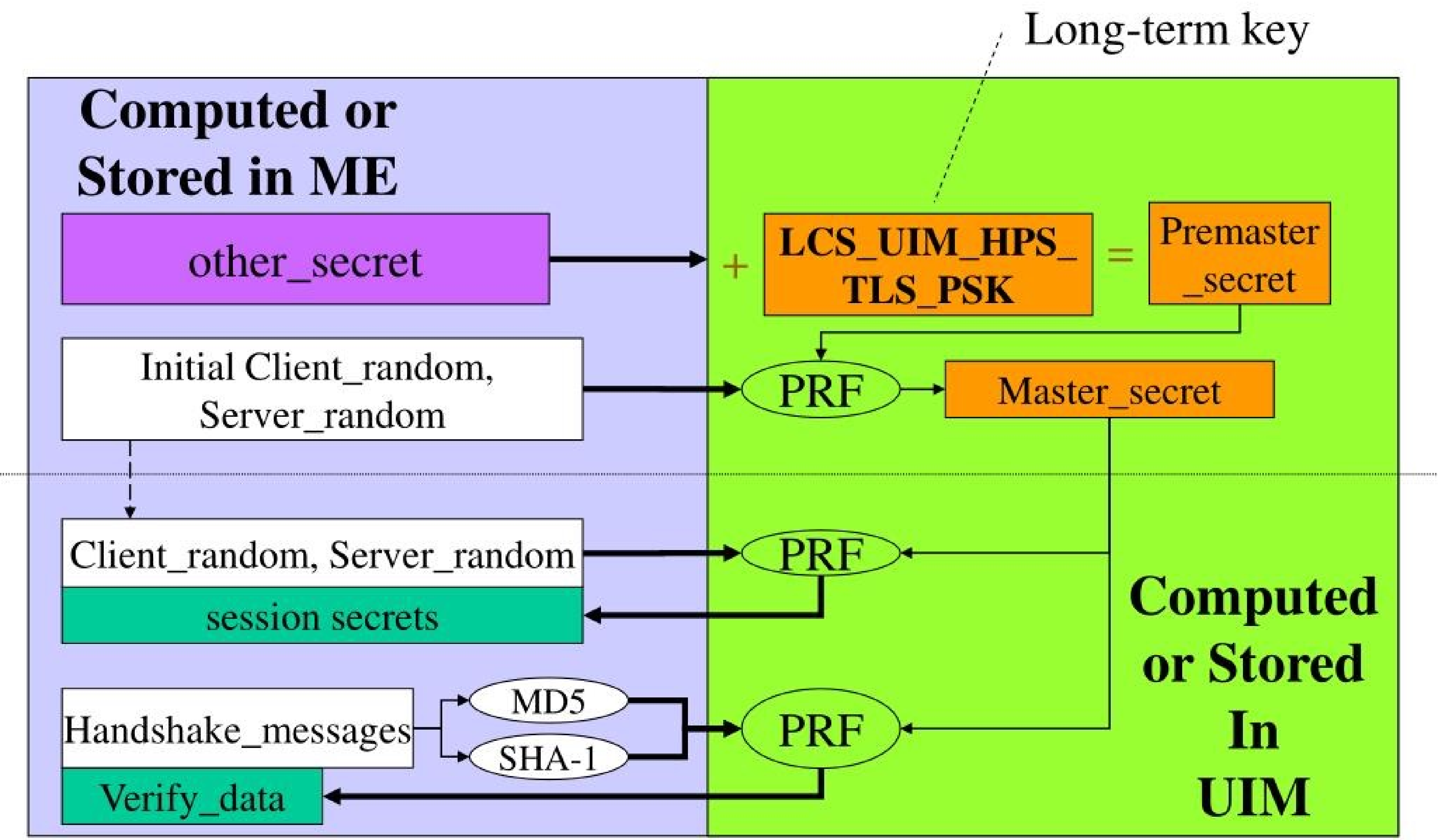
Session Secret Generation Model



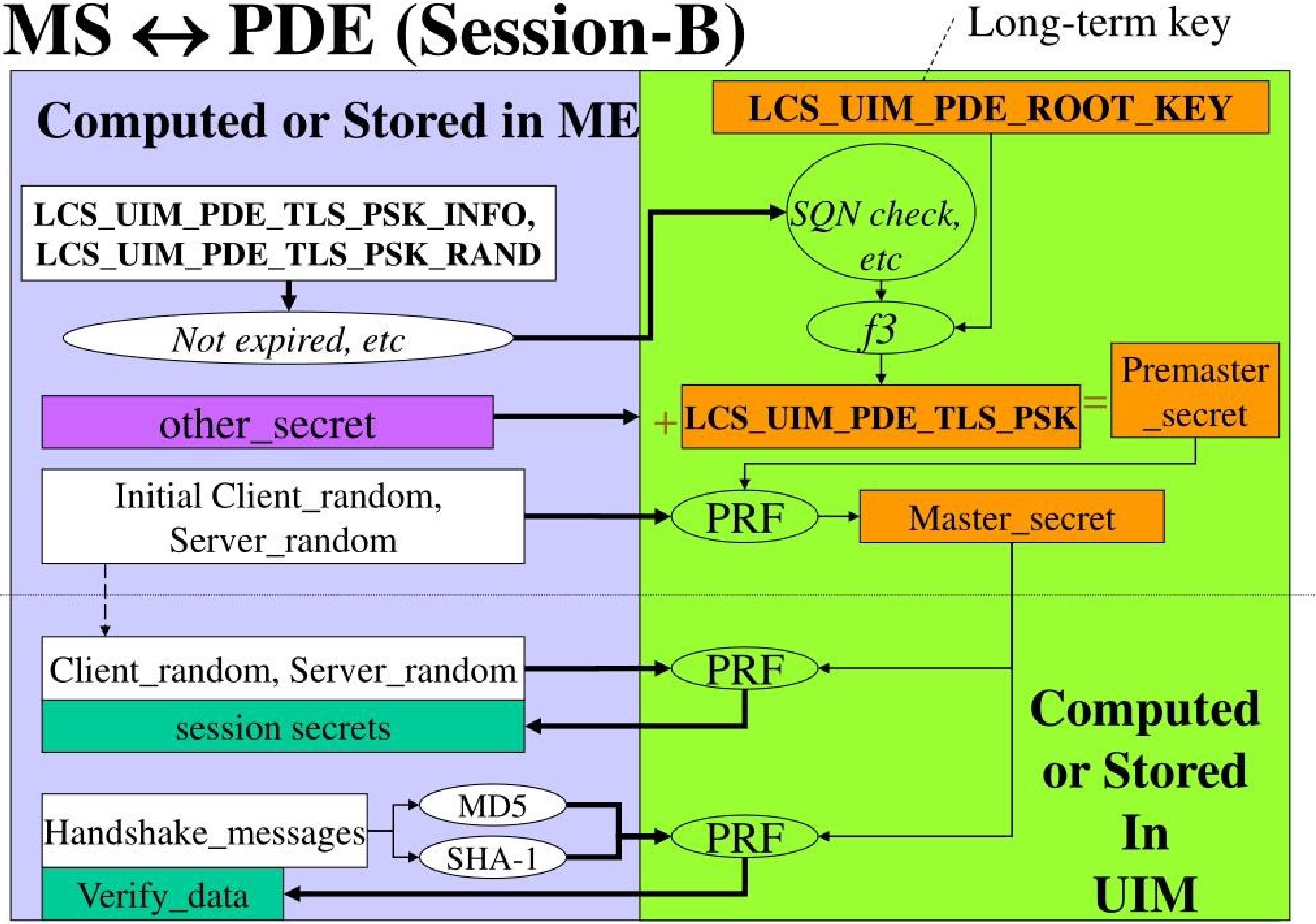
Notes

- UIM doesn't care if pure PSK, or RSA or DHE etc is added.
- UIM must be present for
  - Session secret generation and
  - Verify_data generation
  - Resuming a TLS session
- PSK may be re-used for multiple TLS sessions

MS ↔ H-PS (Session-A)

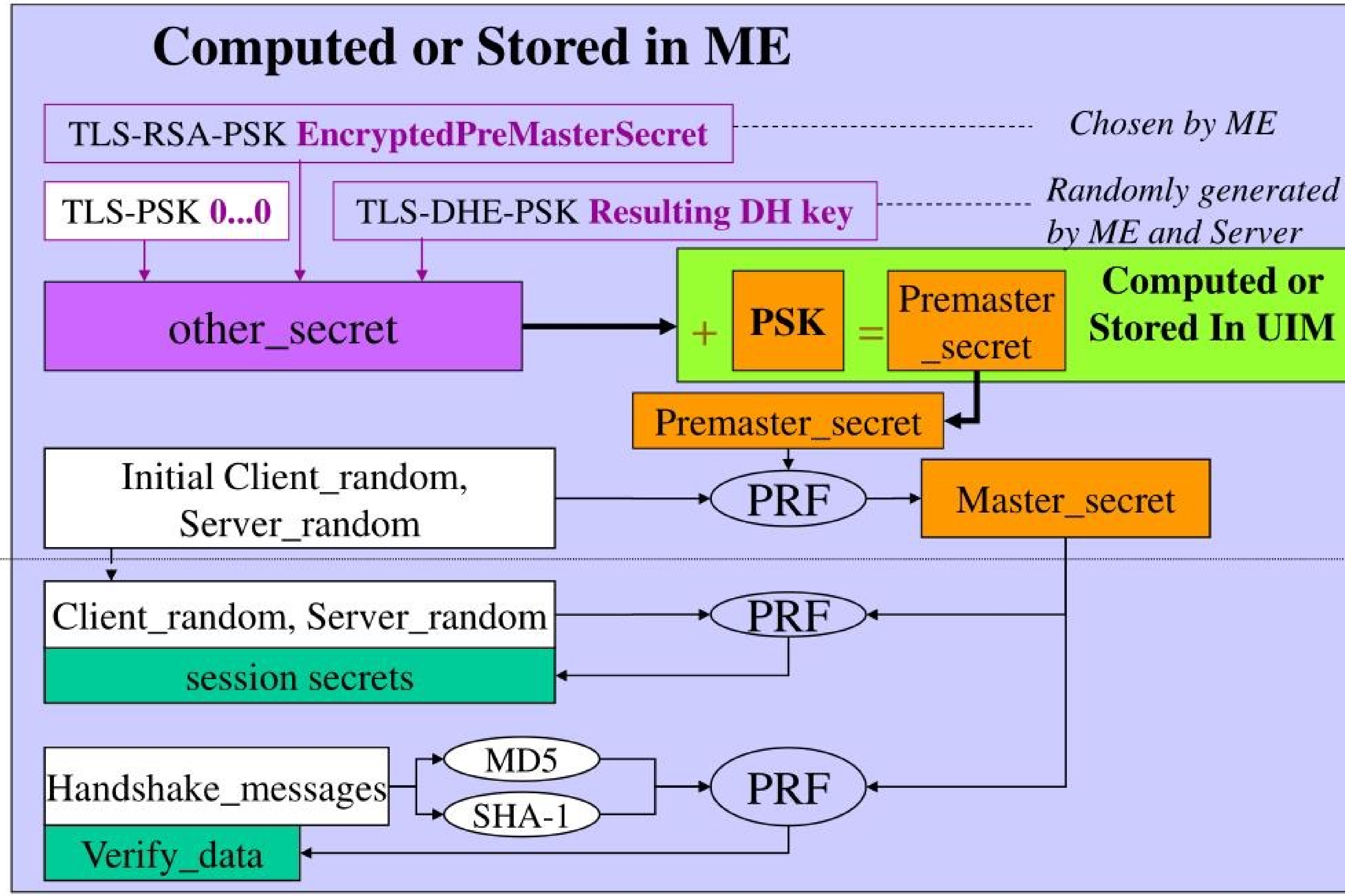


MS ↔ PDE (Session-B)



Why session secrets are generated in UIM

Model 1 (Very BAD)



Problems with Model 1

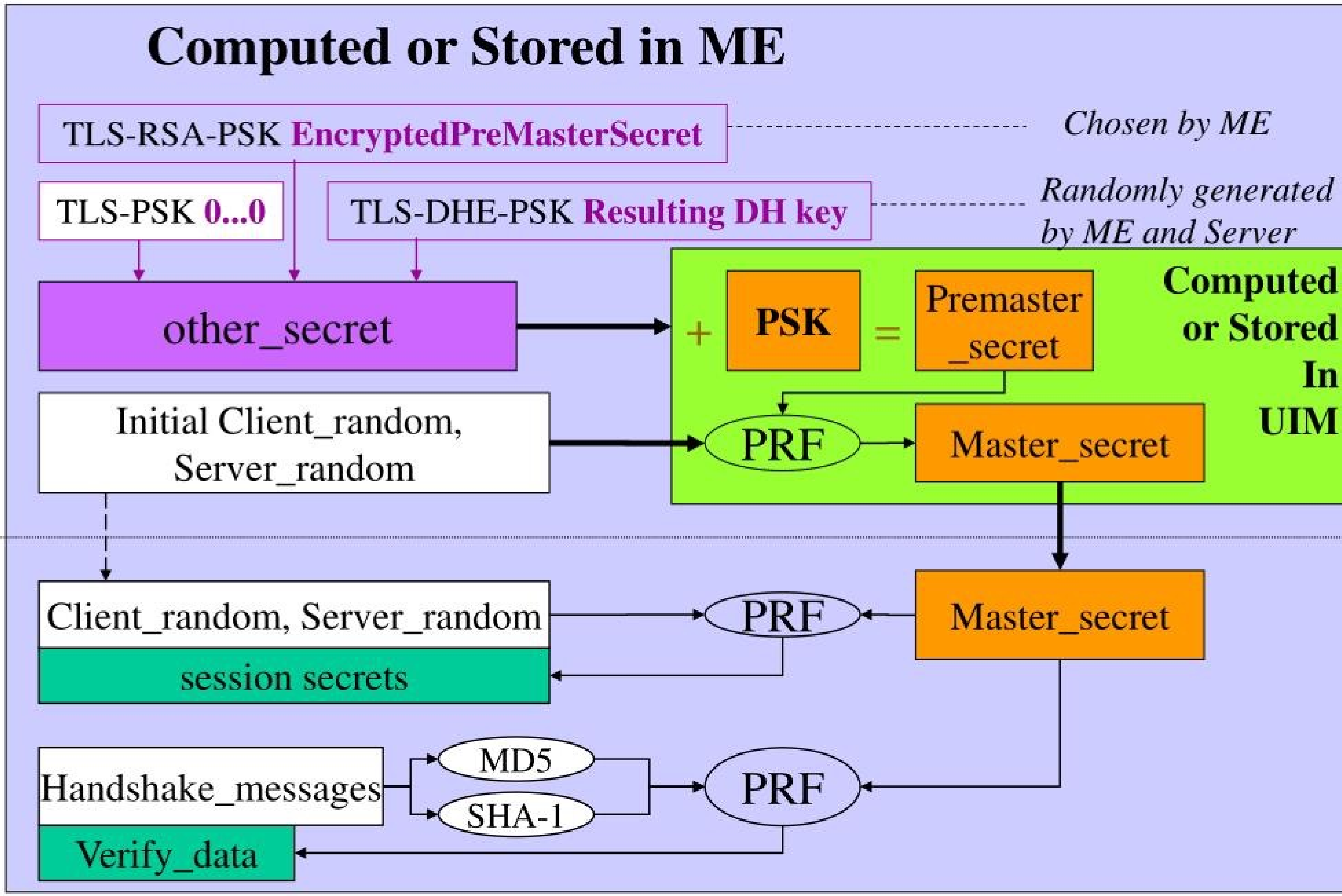
ME Handshake abilities:

- ME can steal and distribute
  - Master_secret (only used in single TLS session)
  - Premaster_secret (can use in multiple TLS sessions)

Consequences if PSK used more than once:

- Other users can start new TLS sessions: **YES**
- Other users can resume this TLS session: **YES**

Model 2 (BAD)



Computed or Stored in ME
TLS-RSA-PSK EncryptedPreMasterSecret
Chosen by ME
TLS-PSK 0...0
TLS-DHE-PSK Resulting DH key
Randomly generated by ME and Server
other_secret
+
PSK
=
Premaster _secret
Computed or Stored In UIM
Initial Client_random, Server_random
PRF
Master_secret
Client_random, Server_random
PRF
Master_secret
session secrets
Handshake_messages
MD5
SHA-1
PRF
Verify_data

Model 2 Analysis

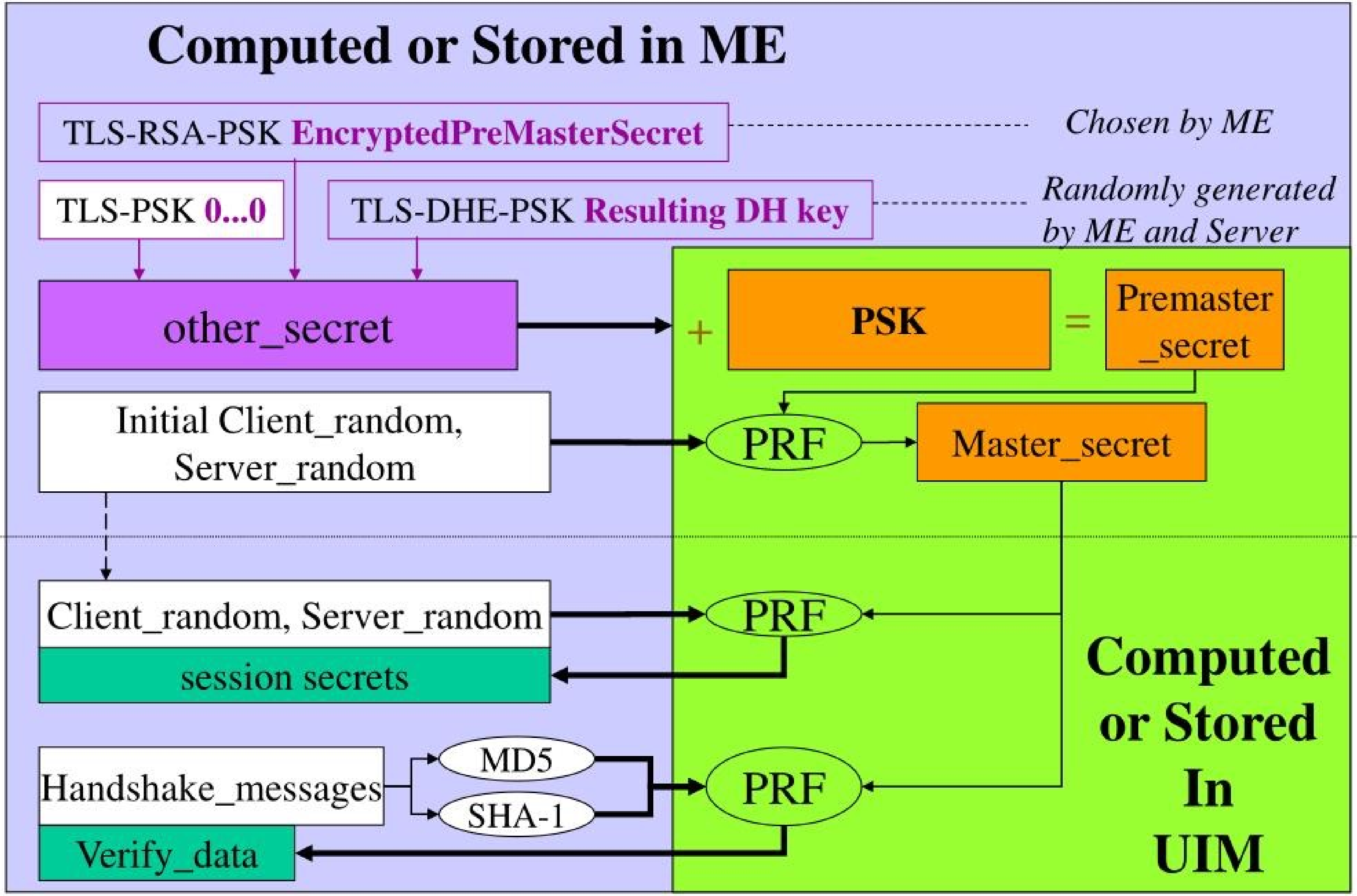
ME Handshake abilities:

- ME can steal and distribute
  - Master_secret (only used in single TLS session)

Consequences if PSK used more than once:

- Other users can start new TLS sessions: **NO**
- Other users can resume this TLS session: **YES**

Model 3 (GOOD)



Computed or Stored in ME
TLS-RSA-PSK EncryptedPreMasterSecret
Chosen by ME
TLS-PSK 0...0
TLS-DHE-PSK Resulting DH key
Randomly generated by ME and Server
other_secret
+
PSK
=
Premaster _secret
Initial Client_random, Server_random
PRF
Master_secret
Client_random, Server_random
PRF
session secrets
Handshake_messages
MD5
SHA-1
PRF
Verify_data
Computed or Stored In UIM

Model 3 Analysis

ME Handshake abilities:

- None

Consequences if PSK used more than once:

- Other users can start new TLS sessions: **NO**
- Other users can resume this TLS session: **NO**